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Boundary Value Analysis (BVA) — Test Design Sheet

1. Vehicle Speed Limit (Smart Traffic System)

S. No.	Boundary Type	Test Data (Condition)	Valid Range	Expected Status
1	Minimum Boundary	20 km/h	20 km/h – 120 km/h	Passed
2	Below Minimum	19 km/h	20 km/h – 120 km/h	Failed
3	Just Above Minimum	21 km/h	20 km/h – 120 km/h	Passed
4	Maximum Boundary	120 km/h	20 km/h – 120 km/h	Passed
5	Just Below Maximum	119 km/h	20 km/h – 120 km/h	Passed
6	Above Maximum	121 km/h	20 km/h – 120 km/h	Failed

2. Water Tank Level (IoT Smart Tank)

Valid Range: 10 L – 100 L

S. No.	Boundary Type	Test Data (Condition)	Valid Range	Expected Status
1	Minimum Boundary	10 L	10 L – 100 L	Passed
2	Below Minimum	9 L	10 L – 100 L	Failed
3	Just Above Minimum	11 L	10 L – 100 L	Passed
4	Maximum Boundary	100 L	10 L – 100 L	Passed
5	Just Below Maximum	99 L	10 L – 100 L	Passed
6	Above Maximum	101 L	10 L – 100 L	Failed

3. Online Exam Duration

Valid Range: 30 min – 180 min

S. No.	Boundary Type	Test Data (Condition)	Valid Range	Expected Status
1	Minimum Boundary	30 min	30 min – 180 min	Passed
2	Below Minimum	29 min	30 min – 180 min	Failed
3	Just Above Minimum	31 min	30 min – 180 min	Passed
4	Maximum Boundary	180 min	30 min – 180 min	Passed
5	Just Below Maximum	179 min	30 min – 180 min	Passed

4. Product Discount Percentage (E-commerce)

Valid Range: 5% – 70%

S. No.	Boundary Type	Test Data (Condition)	Valid Range	Expected Status
1	Minimum Boundary	5%	5% – 70%	Passed
2	Below Minimum	4%	5% – 70%	Failed
3	Just Above Minimum	6%	5% – 70%	Passed
4	Maximum Boundary	70%	5% – 70%	Passed
5	Above Maximum	71%	5% – 70%	Failed

5. Warehouse Temperature (Cold Storage)

Valid Range: -5°C – 8°C

S. No.	Boundary Type	Test Data (Condition)	Valid Range	Expected Status
1	Minimum Boundary	-5°C	-5°C – 8°C	Passed
2	Below Minimum	-6°C	-5°C – 8°C	Failed
3	Just Above Minimum	-4°C	-5°C – 8°C	Passed
4	Maximum Boundary	8°C	-5°C – 8°C	Passed
5	Just Below Maximum	7°C	-5°C – 8°C	Passed
6	Above Maximum	9°C	-5°C – 8°C	Failed

Equivalence Class Partitioning (ECP) — Test Design Sheet

1. File Upload Size

Valid Range: 1 MB – 25 MB

Test Case	Equivalence Class	Test Data	Valid Range	Expected Result
EC1	Valid	10 MB	1 MB – 25 MB	Passed
EC2	Invalid (< 1 MB)	0.5 MB	1 MB – 25 MB	Fail
EC3	Invalid (> 25 MB)	30 MB	1 MB – 25 MB	Fail

2. Product Rating

Valid Range: 1 – 5 Stars

Test Case	Equivalence Class	Test Data	Valid Range	Expected Result
EC1	Valid	4	1 – 5 Stars	Passed
EC2	Invalid (< 1)	0	1 – 5 Stars	Fail
EC3	Invalid (> 5)	6	1 – 5 Stars	Fail

3. Coupon Discount

Valid Range: 5% – 50%

Test Case	Equivalence Class	Test Data	Valid Range	Expected Result
EC1	Valid	25%	5% – 50%	Passed
EC2	Invalid (< 5%)	3%	5% – 50%	Fail
EC3				Fail

4. Event Registration Capacity

Valid Range: 50 – 500 Participants

Test Case	Equivalence Class	Test Data	Valid Range	Expected Result
EC1	Valid	250	50 – 500 Participants	Passed
EC2				Fail
EC3	Invalid (> 500)	550	50 – 500 Participants	Fail

5. UPI Transaction Amount

Valid Range: ₹1 – ₹1,00,000

Test Case	Equivalence Class	Test Data	Valid Range	Expected Result
EC1	Valid	₹25,000	₹1 – ₹1,00,000	Passed
EC2	Invalid (< ₹1)	₹0	₹1 – ₹1,00,000	Fail
EC3	Invalid (> ₹1,00,000)	₹1,50,000	₹1 – ₹1,00,000	Fail